MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 02
Chapter 11: Areas Related to Circles

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MULTIPLE CHOICE

MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. The length of an arc of a sector of angle θ (in degrees) of a circle with radius r is: [1]

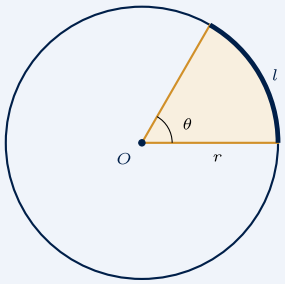
(A) $\frac{\theta}{180^\circ} \times 2\pi r$ (B) $\frac{\theta}{360^\circ} \times 2\pi r$ (C) $\frac{\theta}{180^\circ} \times \pi r^2$ (D) $\frac{\theta}{360^\circ} \times \pi r^2$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Arc Length Formula Derivation [0.5 Mark]

The entire circumference $2\pi r$ corresponds to a complete angle of 360°. For a central angle θ, the arc length l is proportional:

$$l = \frac{\theta}{360^\circ} \times 2\pi r \quad \left(\text{or } \frac{\theta}{180^\circ} \times \pi r\right)$$



Step 2: Matching Options [0.5 Mark]

Option (B) represents the exact standard formula.

Correct Option: (B) $\frac{\theta}{360^\circ} \times 2\pi r$

$$\boxed{\text{Arc Length } l = \frac{\theta}{\mathbf{360^\circ}} \times \mathbf{2\pi r}}$$

Q2. The area of a sector of angle p° of a circle with radius R is: [1]

(A) $\frac{p}{180} \times 2\pi R$ (B) $\frac{p}{180} \times \pi R^2$ (C) $\frac{p}{360} \times 2\pi R$ (D) $\frac{p}{720} \times 2\pi R^2$

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Standard Area of Sector Formula [0.5 Mark]

$$\text{Area of sector} = \frac{\theta}{360} \times \pi R^2 = \frac{p}{360} \times \pi R^2$$

Step 2: Equivalent Expression [0.5 Mark]

Multiplying both numerator and denominator by 2:

$$\frac{p}{360} \times \pi R^2 = \frac{p \times 2}{360 \times 2} \times \pi R^2 = \frac{p}{720} \times (2\pi R^2)$$

Correct Option: (D) $\frac{p}{720} \times 2\pi R^2$

Correct Option: (**D**)

Q3. If the area of a sector of a circle bounded by an arc of length 5π cm is equal to 20π cm^2, then the radius of the circle is: [1]

(A) 12 cm (B) 16 cm (C) 8 cm (D) 10 cm

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Direct Relation between Sector Area and Arc Length [0.5 Mark]

$$\text{Area of sector } A = \frac{1}{2}lr$$

Given $l = 5\pi$ cm and $A = 20\pi$ cm^2.

Step 2: Solving for Radius r [0.5 Mark]

$$20\pi = \frac{1}{2} \times (5\pi) \times r \implies 20 = \frac{5}{2}r \implies r = \frac{20 \times 2}{5} = 8 \text{ cm}$$

Correct Option: (C) 8 cm

Radius of the circle $r = \mathbf{8}$ cm

Q4. If the area of a sector of a circle is $\frac{5}{18}$ of the area of the circle, then the central angle of the sector is: [1]

(A) 60° (B) 90° (C) 100° (D) 120°

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Setting up the Proportion [0.5 Mark]

$$\text{Area of sector} = \frac{\theta}{360^\circ} \times \pi r^2 = \frac{5}{18} \times \pi r^2$$

Step 2: Solving for θ [0.5 Mark]

$$\frac{\theta}{360^\circ} = \frac{5}{18} \implies \theta = \frac{5}{18} \times 360^\circ = 5 \times 20^\circ = 100^\circ$$

Correct Option: (C) 100°

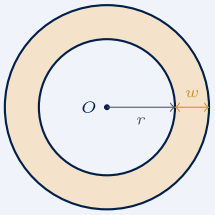
Central Angle $\theta = \mathbf{100}^\circ$

Q5. **Assertion (A):** A running track is in the form of a circular ring. If the inner and outer circumferences are 352 m and 396 m respectively, then the width of the track is 7 metres.
Reason (R): The width of a circular annulus with outer circumference C_2 and inner circumference C_1 is given by $w = \dfrac{C_2 - C_1}{2\pi}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Verification of Reason (R) [0.5 Mark]



Let outer radius be R and inner radius be r. Then $C_2 = 2\pi R \implies R = \frac{C_2}{2\pi}$ and $C_1 = 2\pi r \implies r = \frac{C_1}{2\pi}$.

$$\text{Width } w = R - r = \frac{C_2 - C_1}{2\pi}$$

Hence Reason (R) is **True**.

Step 2: Verification of Assertion (A) [0.5 Mark]

$$w = \frac{396 - 352}{2 \times \frac{22}{7}} = \frac{44 \times 7}{44} = 7 \text{ m}$$

Hence Assertion (A) is **True**, and (R) correctly explains (A).
Correct Option: (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

Correct Option: (**A**)

SECTION B: Very Short Answer Type — MATHS MANTHAN | (1 × 2 = 2 Marks)

Q6. The circumference of a circle exceeds its diameter by 16.8 cm. Find the circumference of the circle. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Formulate Algebraic Equation for Radius r [1 Mark]

Given: Circumference − Diameter = 16.8 cm

$$2\pi r - 2r = 16.8 \implies 2r(\pi - 1) = 16.8$$

$$2r\left(\frac{22}{7} - 1\right) = 16.8 \implies 2r\left(\frac{15}{7}\right) = 16.8$$

$$2r = \frac{16.8 \times 7}{15} = \frac{117.6}{15} = 7.84 \text{ cm} \implies r = \frac{7.84}{2} = 3.92 \text{ cm}$$

Step 2: Compute Circumference [1 Mark]

$$\text{Circumference} = 2\pi r = 2 \times \frac{22}{7} \times 3.92 = 44 \times 0.56 = 24.64 \text{ cm}$$

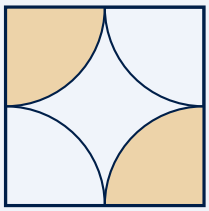
Circumference of the circle = **24.64** cm

SECTION C: Short Answer Type — MATHS MANTHAN | (3 × 3 = 9 Marks)

Q7. Find the area of the shaded region in a square $ABCD$ of side 14 cm, where four semicircles are drawn with each side of the square as diameter. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Side of square $a = 14$ cm. Diameter of each semicircle $d = 14$ cm $\implies$ $r = 7$ cm.



Step 1: Deconstruct into Opposite Semicircles [1 Mark]

Let the four unshaded regions inside the square be I, II, III, and IV.

$$\text{Area of square } ABCD = 14 \times 14 = 196 \text{ cm}^2$$

Area of (I + III) = Area of square − Area of 2 semicircles on AD and BC

$$= 196 - 2 \times \left(\frac{1}{2}\pi r^2\right) = 196 - \frac{22}{7} \times 49 = 196 - 154 = 42 \text{ cm}^2$$

Step 2: Total Unshaded Region [1 Mark]

Similarly, Area of (II + IV) = 42 cm^2.

$$\text{Total unshaded area (I + II + III + IV)} = 42 + 42 = 84 \text{ cm}^2$$

Step 3: Area of Shaded Region (4 Petals) [1 Mark]

$$\text{Area of shaded region} = \text{Area of square} - \text{Total unshaded area}$$
$$= 196 - 84 = 112 \text{ cm}^2$$

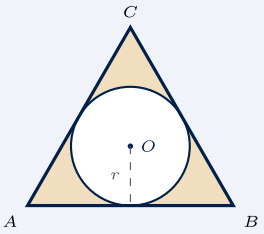
Area of Shaded Region = **112 cm^2**

Q8. An equilateral triangle of side 12 cm has an inscribed circle touching its sides. Find the area of the region inside the triangle but outside the circle. (Use $\pi = 3.14, \sqrt{3} = 1.732$) **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Side of equilateral triangle $a = 12$ cm.

Step 1: Area of Equilateral Triangle [1 Mark]



$$\text{Area}(\triangle ABC) = \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4} \times (12)^2 = 36\sqrt{3} \text{ cm}^2$$
$$= 36 \times 1.732 = 62.352 \text{ cm}^2$$

Step 2: Radius and Area of Inscribed Circle [1 Mark]

For an equilateral triangle of side a, inradius r:

$$r = \frac{a}{2\sqrt{3}} = \frac{12}{2\sqrt{3}} = 2\sqrt{3} \text{ cm}$$

$$\text{Area of Incircle} = \pi r^2 = 3.14 \times (2\sqrt{3})^2 = 3.14 \times 12 = 37.68 \text{ cm}^2$$

Step 3: Area of Region Outside Circle [1 Mark]

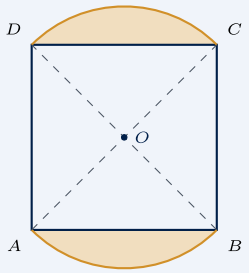
$$\text{Area} = \text{Area}(\triangle ABC) - \text{Area of Incircle}$$
$$= 62.352 - 37.68 = 24.672 \text{ cm}^2 \approx 24.67 \text{ cm}^2$$

Area of Region Outside Circle = **24.67 cm^2**

Q9. Two circular flower beds are set up on two opposite sides of a square lawn $ABCD$ of side 56 m. The centre of each circular flower bed is the intersection point O of the diagonals of the lawn. Find the sum of the areas of the lawn and the flower beds. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Side of square lawn $a = 56$ m. Let O be the intersection of diagonals AC and BD.



Step 1: Radius Squared (r^2) [1 Mark]

In right $\triangle ABC$, $AC^2 = 56^2 + 56^2 = 2 \times 56^2$.

$$OA = OB = r = \frac{AC}{2} \implies r^2 = \frac{AC^2}{4} = \frac{2 \times 56^2}{4} = \frac{3136}{2} = 1568$$

Step 2: Area Decomposition [1 Mark]

Diagonals of a square are perpendicular bisectors ($\angle AOB = \angle COD = 90°$).

$$\text{Area of sector } OAB = \frac{90°}{360°} \times \pi r^2 = \frac{1}{4} \times \frac{22}{7} \times 1568 = \frac{11}{14} \times 1568 = 1232 \text{ m}^2$$

$$\text{Similarly, Area of sector } OCD = 1232 \text{ m}^2$$

Step 3: Total Area of Lawn and Flower Beds [1 Mark]

$$\text{Area of } \triangle OBC = \frac{1}{4} \times \text{Area of square} = \frac{1}{4} \times 56 \times 56 = 784 \text{ m}^2$$

$$\text{Area of } \triangle OAD = 784 \text{ m}^2$$

$$\text{Total Area} = \text{Area}(OAB) + \text{Area}(OCD) + \text{Area}(\triangle OBC) + \text{Area}(\triangle OAD)$$

$$= 1232 + 1232 + 784 + 784 = 2464 + 1568 = 4032 \text{ m}^2$$

$$\boxed{\text{Total Area of Lawn and Flower Beds} = \mathbf{4032} \text{ m}^2}$$

SECTION D: Case-Based / Applied Situation

MATHS MANTHAN | (1 × 4 = 4 Marks)

Q10. Case Study: Handicraft Block Printing on Silk Handkerchief

A textile artisan designs a traditional block print on a square silk handkerchief of side 42 cm. The pattern consists of nine identical circular motifs arranged neatly in three rows of three circles each, touching the adjacent circles and handkerchief edges. Each circle has radius $r = 7$ cm.

(i) Calculate the total area of the square silk handkerchief. **[1]**

(ii) Calculate the combined area occupied by all nine circular motifs. **[1]**

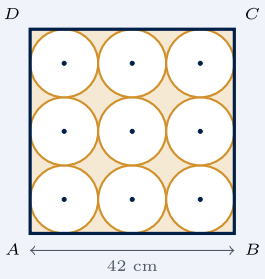
(iii) Find the remaining unprinted area of the silk handkerchief. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[2]**

— OR —

If each circular motif costs Rs. 5 to print and the remaining area is dyed at Rs. 0.05 per cm^2, find the total production cost. **[2]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Given: Side of square handkerchief $a = 42$ cm. Radius of each circle $r = 7$ cm.



Part (i): Area of Square Handkerchief [1 Mark]

$$\text{Area of square} = a^2 = 42 \times 42 = 1764 \text{ cm}^2$$

Part (ii): Combined Area of 9 Circular Motifs [1 Mark]

$$\text{Area of one circular motif} = \pi r^2 = \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$$

$$\text{Area of 9 motifs} = 9 \times 154 = 1386 \text{ cm}^2$$

Part (iii): Remaining Unprinted Area [2 Marks]

$$\text{Remaining unprinted area} = \text{Area of square} - \text{Area of 9 motifs}$$

$$= 1764 - 1386 = 378 \text{ cm}^2$$

— OR (Alternative Choice) —

Total Production Cost Calculation: [2 Marks]

$$\text{Printing cost for 9 circular motifs} = 9 \times \text{Rs. } 5 = \text{Rs. } 45$$

$$\text{Dyeing cost for remaining area} = 378 \times \text{Rs. } 0.05 = \text{Rs. } 18.90$$

$$\text{Total Production Cost} = 45 + 18.90 = \text{Rs. } 63.90$$

(i) 1764 cm^2, (ii) 1386 cm^2, (iii) Remaining Area = **378** cm^2 (OR Total Cost = Rs. **63.90**)

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